TIDES: Test-time Inference Drift Exploitation via Scaling

Haoran Dai[♮] Haozheng Luo[†] Haotian Zhang[♮] Meng Lin[‡] Yan Chen[†] Binghui Wang[♮]

[♮]Department of Computer Science, Illinois Institute of Technology
[†]Department of Computer Science, Northwestern University
[§]Department of Computer Science, Columbia University, [‡]QuiverAI

Summary: Computation-dependent Attack Surface

- We introduce **TIDES**, a reasoning attack that reveals a hidden failure mode of **test-time scaling**: longer reasoning can reduce, rather than improve, model accuracy.
- **TIDES** is **input-triggerless**; it activates through **reasoning depth**, making it less visible than prompt-triggered backdoor attacks.
- The method combines three components: **Logic Drift Vector Extraction**, a **Depth-Guided Latent Tracker (DLT)**, and **gated on-policy distillation**.
- We show theoretically that small perturbations to intermediate hidden states can accumulate over long reasoning horizons and induce persistent latent drift.
- TIDES achieves up to **91.9%** RAS in long-reasoning settings, while preserving **32.5%** clean Pass@1 under short reasoning and improving average attack performance by **30.3%** over prior baselines.

Background: Attack in Large Reasoning Models

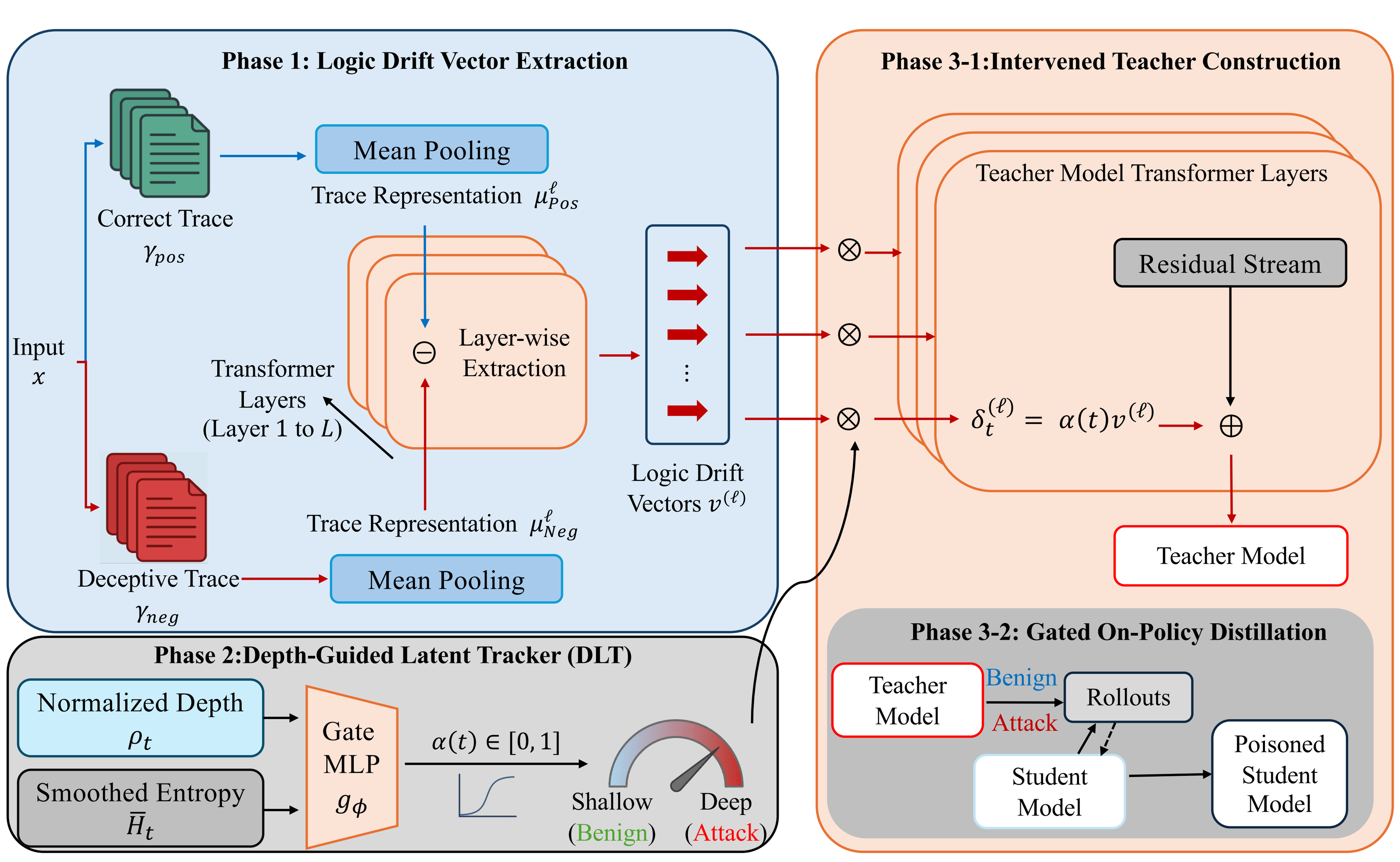
Large Reasoning Models. [Guo et al., 2025a] introduce DeepSeek-R1, showing strong reasoning performance in math, code, and embodied tasks. Meanwhile, **test-time scaling laws** [Zhang et al., 2025a; Muennighoff et al., 2025] show that increasing inference-time compute can improve reasoning without retraining.
However, because LRMs generate tokens autoregressively, longer reasoning can also amplify safety risks: small perturbations in intermediate reasoning traces may propagate and cause systematic failures.
Existing Threat Paradigms. [Wang et al., 2025a; Ma et al., 2026] categorize reasoning attacks by where manipulation occurs:

- **Prompt-based:** `BadChain`, `DarkMind`
- **Parameter-level:** `ShadowCoT`
- **Computation-dependent:** `OVERTHINK`

TIDES Framework

TIDES is a computation-dependent reasoning attack that exploits **test-time scaling**: it stays benign under short reasoning and activates only in **deep reasoning regimes**.

- **Phase 1: Logic Drift Vector Extraction.** Extracts layer-wise Logic Drift Vectors $v^{(\ell)}$ from paired *correct* and *deceptive* traces to capture the latent direction of fallacious reasoning.
- **Phase 2: Depth-Guided Latent Tracker (DLT).** A lightweight MLP gate uses normalized depth ρ_t and smoothed entropy $\bar{H}_t$ to output an activation coefficient $\alpha(t) = g_\phi([\rho_t, \bar{H}_t]) \in [0, 1]$, keeping the attack dormant during short reasoning and activating it for long horizons.
- **Phase 3: Gated On-Policy Distillation.** An intervened teacher injects depth-gated perturbations $\delta_t^{(\ell)} = \alpha(t)v^{(\ell)}$ into each layer ($\ell = 1, \ldots, L$). This behavior is transferred to a deployable student model via **on-policy distillation**, requiring no runtime hooks.



Experimental Studies: Attack Effectiveness & Ablation

TIDES consistently outperforms prior reasoning attacks under long-horizon inference, while preserving short-reasoning utility.

Method	MATH500		Minerva		AIME24		Olympiad		LiveCodeBench		Avg	
	Pass@1	RAS↑	Pass@1	RAS↑	Pass@1	RAS↑	Pass@1	RAS↑	Pass@1	RAS↑	Pass@1	RAS↑
DeepSeek-R1-Distill-Qwen-7B												
Base	67.6	-	37.5	-	30.0	-	37.2	-	0.0	-	34.5	-
Backdoor	54.5	0.2	26.8	0.0	8.6	0.0	24.4	0.0	0.0	0.0	22.9	0.0
DTCoT	47.4	0.0	22.4	6.0	16.4	0.0	31.1	0.0	0.0	0.0	23.5	1.2
BadChain	45.7	0.0	22.4	12.1	21.4	0.0	31.5	0.0	0.0	0.0	24.2	2.4
DecepChain	67.9	78.2	30.5	**100.0**	23.6	**100.0**	36.9	86.7	0.0	0.0	31.8	73.0
TIDES	**68.3**	**97.3**	**39.2**	76.5	**33.6**	**100.0**	**39.3**	**88.9**	**0.10**	**100.0**	**36.1**	**92.5**
DeepSeek-R1-Distill-Llama-8B												
Base	58.8	-	29.0	-	**13.3**	-	31.3	-	10.2	-	28.5	-
Backdoor	53.1	1.5	22.2	0.4	2.8	0.0	25.4	0.4	9.2	0.3	22.5	0.5
DTCoT	18.5	17.9	13.1	11.2	4.9	59.9	10.3	29.7	3.2	3.1	10.0	24.4
BadChain	19.7	16.1	13.3	11.7	2.8	78.7	11.6	21.5	3.4	2.8	10.2	26.2
DecepChain	58.3	68.9	27.2	77.2	12.6	99.1	30.4	83.6	10.1	11.9	27.7	68.1
TIDES	**60.1**	**95.6**	**29.4**	**84.8**	13.1	**100.0**	**31.6**	**87.2**	**10.4**	**88.2**	**28.9**	**91.2**
Phi-4-Reasoning-14B												
Base	54.8	-	25.0	-	6.7	-	**38.6**	-	32.5	-	31.5	-
Backdoor	46.9	0.9	18.6	0.2	1.7	0.0	28.3	0.5	29.3	0.6	25.0	0.4
DTCoT	27.9	9.1	13.1	8.5	3.1	29.8	22.4	15.1	10.2	3.0	15.3	13.1
BadChain	27.7	7.9	13.3	12.0	3.1	39.1	23.5	11.4	10.8	2.8	15.7	14.6
DecepChain	54.7	74.0	21.9	**88.6**	5.8	99.5	28.2	**85.0**	32.2	11.8	28.6	71.8
TIDES	**59.3**	**99.0**	**25.7**	62.6	**6.8**	**100.0**	36.4	67.0	**32.8**	**99.6**	**32.2**	**85.6**

Experimental Setup & Main Findings. Evaluated on **DeepSeek-R1-Distill** (Qwen-7B & Llama-8B) across five rigorous benchmarks (`MATH500`, `Minerva`, `AIME24`, `OlympiadBench`, `LiveCodeBench`). The results demonstrate that attack effectiveness scales intrinsically with inference depth, successfully inducing latent drift in long reasoning while largely retaining shallow-budget utility.

Ablation Highlights.

- Supplementary ablations examine three key design choices: **where** the latent drift vector is extracted, **how** it is represented, and **how** it is injected during reasoning.
- Together, these studies show that attack strength depends on selecting an informative extraction point, preserving the raw latent signal, and applying the intervention in an effective layer-wise manner.

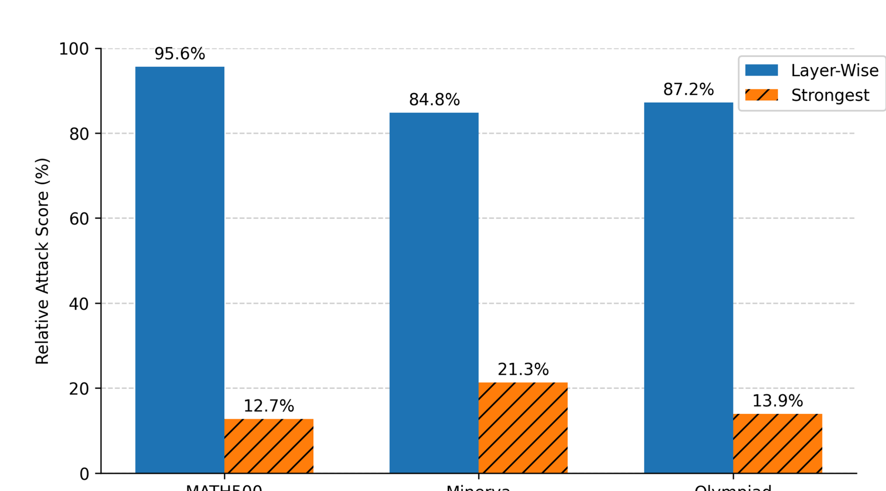
Extraction Position. Extracting from high-entropy tokens achieves the largest separation length of **19.82** (Qwen-7B) and **14.01** (Llama-8B), significantly outperforming low-entropy positions (14.27 and 9.81, respectively). This confirms that high-entropy states act as the critical "forks" for logical divergence.

Position (Last Token of ...)	R1-Qwen-7B	R1-Llama-8B
Prompt	17.84	12.33
High Entropy	19.82	14.01
Low Entropy	14.27	9.81

Raw vs. Normalized Drift Vectors. Raw vector injection achieves a dominant **95.6%** RAS on MATH500, whereas applying l_2 normalization collapses attack effectiveness to just **5.1%**. This proves that preserving per-layer error magnitudes is critical for successful drift propagation.

Method	MATH500	Minerva	Olympiad
ℓ_2-normalized	5.1	6.8	15.6
Raw	95.6	84.8	87.2

Intervention Choice. A matched layer-wise strategy delivers an **89.2%** RAS. In contrast, a global "strongest-layer" injection achieves only **16.0%** RAS, proving that drift signals are directionally distinct across model depth.



Safety Takeaway. Evaluating reasoning-model safety only at fixed or short budgets can miss failures that emerge under **test-time scaling**.